

2.2.2 Two-Valued Boolean Algebra

A two-valued Boolean algebra is defined on a set of two elements, $B = \{0, 1\}$, with rules for the two binary operators $+$ and $\cdot$ as shown in the following operator tables (the rule for the complement operator is for verification of postulate 5):

x	y	$x \cdot y$	$x + y$	$\bar{x}$	$\bar{y}$
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	0	1
1	1	1	1	0	0

These rules are exactly the same as the AND, OR and NOT operations, respectively defined in Table 1-6.

We must now show that the Huntington postulates are valid for the set $B = \{0, 1\}$ and the two binary operators defined above.

1. Closure is obvious from the tables since the result of each operation is either 1 or 0 and $1, 0 \in B$.

2. From the tables we see that: (a) $0+0=0$ (b) $1+1=1$ (c) $0+1=1+0=1$ (d) $1 \cdot 0 = 0 \cdot 1 = 0$ which establishes the two identity elements 0 for $+$ and 1 for $\cdot$ as defined by postulate 2.

3. The commutative laws are obvious from the symmetry of the binary operator tables.

4. (a) The distributive law $x \cdot (y+z) = (x \cdot y) + (x \cdot z)$ can be shown to hold true from the operator tables by forming a truth table of all possible values of x, y , and z .

For each combination, we derive $x \cdot (y+z)$ and show that the value is the same as $(x \cdot y) + (x \cdot z)$.

x	y	z	$(y+z)$	$x \cdot (y+z)$	$(x \cdot y)$	$(x \cdot z)$	$(x \cdot y) + (x \cdot z)$
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1